

Prolog Program to Implement Family Tree

1. Aim

To write and execute a Prolog program to represent a family tree and derive relationships such as father, mother, grandparent, and sibling using facts, rules, and queries.

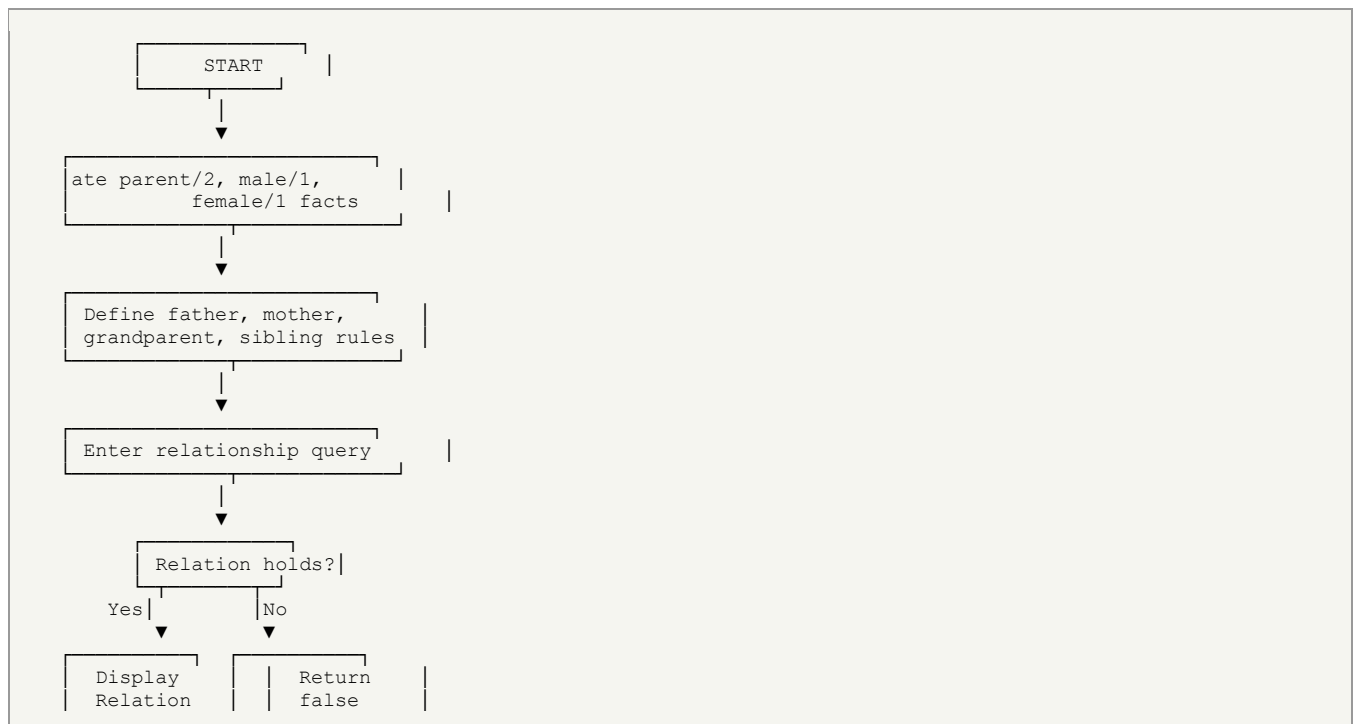
2. Objective

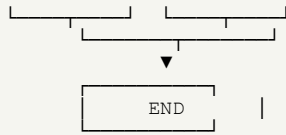
- To understand how a family tree can be represented using base facts (parent, gender).
- To derive higher-level relationships using rules built on top of base facts.
- To retrieve relationships using queries.
- To understand conjunctions of goals within a single rule body.

3. Algorithm

1. Start the program.
2. Create a Prolog database using the predicates parent/2, male/1, and female/1.
3. Store each parent-child pair and each person's gender as facts.
4. Define rules father/2, mother/2, grandparent/2, and sibling/2 built on the base facts.
5. Enter a relationship query, such as grandparent(X, Y).
6. Search the database and apply the relevant rule.
7. If the relationship holds, display the result; otherwise Prolog returns false.
8. Stop the program.

4. Flowchart





5. Code with Output

Code:

```
% -----  
% Family Tree Database  
% -----  
parent(john, mary).  
parent(john, mike).  
parent(susan, mary).  
parent(susan, mike).  
parent(mary, alice).  
parent(mike, bob).  
  
male(john).  
male(mike).  
male(bob).  
  
female(susan).  
female(mary).  
female(alice).  
  
% -----  
% Derived relationships  
% -----  
father(X, Y)      :- parent(X, Y), male(X).  
mother(X, Y)      :- parent(X, Y), female(X).  
grandparent(X, Z) :- parent(X, Y), parent(Y, Z).  
sibling(X, Y)     :- parent(P, X), parent(P, Y), X \= Y.
```

Output:

```
?- father(john, mary).  
true.  
  
?- grandparent(john, alice).  
true.  
  
?- sibling(mary, mike).  
true.
```

6. Result

The Prolog program was successfully implemented to represent a family tree and derive relationships such as father, mother, grandparent, and sibling from base facts.

7. Conclusion

Thus, the family tree program was successfully implemented using Prolog. The experiment demonstrates how complex relationships can be derived from a small set of base facts using compound rule bodies.